

UBL figures: a uniform set

A demonstration for the TC, not a decision. Five of the 78 activity diagrams, from the simplest to the most complex, shown three ways.

1. **As published today.** Each figure is the committed SVG, faithful to its artwork, and shown at the column's full width, as the specification shows every figure. So a figure with a few boxes has large boxes and large text, and a figure with many has small ones.
2. **In a house style, fitted to the column.** The same element looks the same in every figure: one label size, one line weight, one arrowhead, one shape for each kind of node. It is still shown at the column's width, so its size on paper still depends on how much the figure holds: its labels print at exactly the size they do today, since each figure keeps its layout and is scaled as a whole.
3. **In the house style, at one scale for all.** Every figure is printed at the same scale, so a label is the same size on paper (7 pt) in every figure. A simple figure is small; a complex one needs a larger page, or a new layout that fits the page.

In draw.io this is natural. A draw.io page is not a fixed size: the drawing is made in its own units, the same for every figure, and the page and the export scale decide how large it prints. What the TC has to choose is which of the two kinds of uniform it wants on paper: the same *look* (2), or the same *size* as well (3).

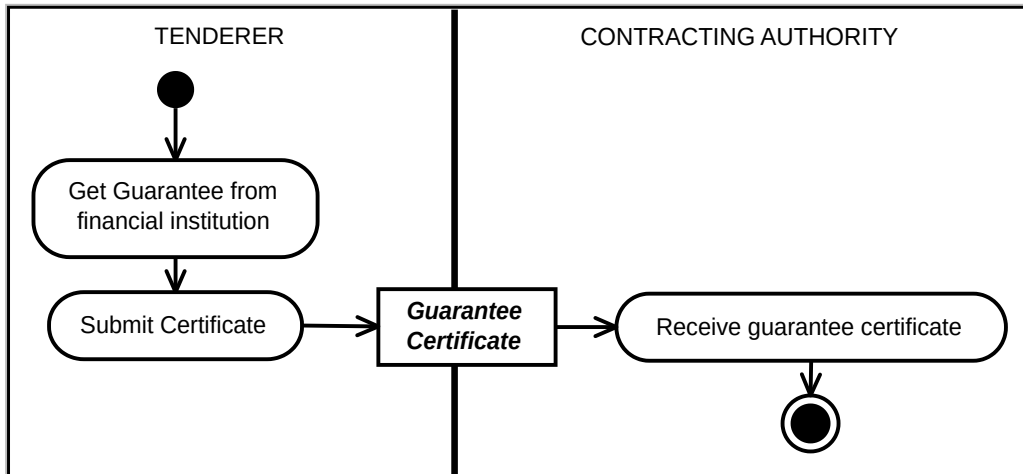
The house style proposed

The median of what the 78 artworks draw, relative to their own label size (poc-drawio/census.md).

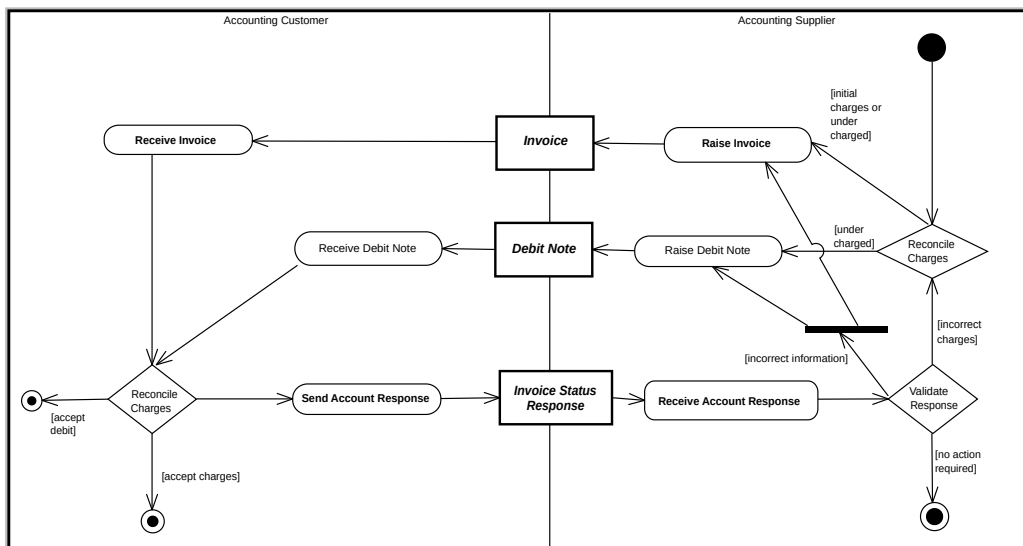
element	house style
labels	12 px Helvetica; actions bold, documents bold italic, decisions, guards and lane titles plain
lines	flows and outlines 0.10 of the label size, documents 0.16, dividers 0.11, frame 0.18
arrowheads	open, 1.35 label sizes long
arrowheads	the same head on every flow
action	its measured size, grown only where its words would not fit; one corner radius in every action (16 px)
document	its measured size, grown only where its words would not fit; centred exactly on the lane divider it sits on
decision	its measured size, grown only around its question; 2 label sizes across when the question stands beside it
lane titles	at the same place in every figure, plain, no rule under them
start, end	draw.io's UML start and end states, 1.8 label sizes across
fork bar	0.5 label sizes thick
grey rules	none: one line per divider

Flows almost level or upright are made exactly so. Each figure keeps its layout: every element stays where its artwork puts it, scaled so its labels come out at the house size; boxes keep their size. Where a box grown to its words comes too close to another, or out of its lane, the figure is opened up there (section 5): over all 78, 19 need nothing, the median figure grows by 1 %, the most by 19 %. The routing of flows is left as drawn (section 4).

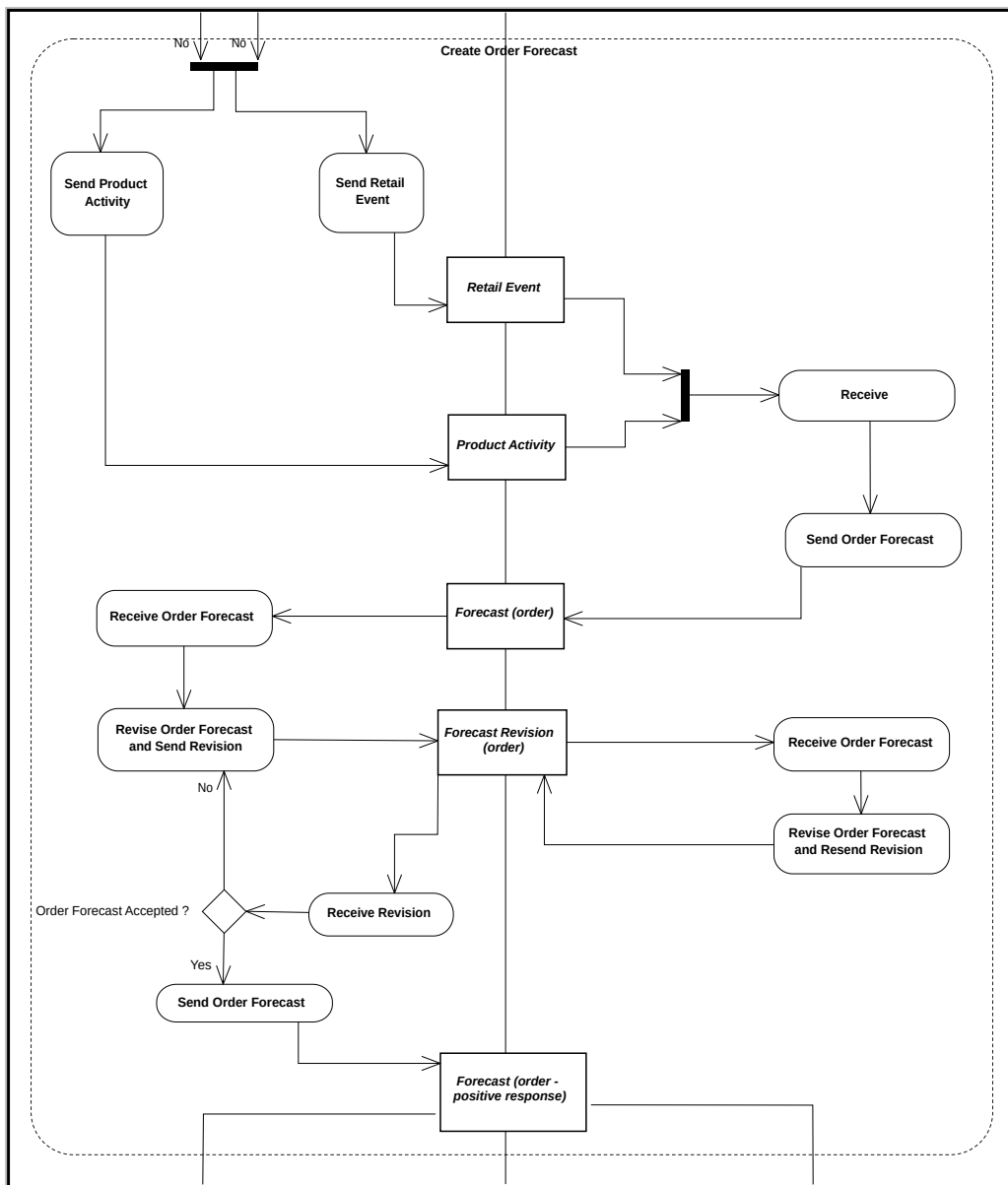
1. As published today: faithful, each at the column's width



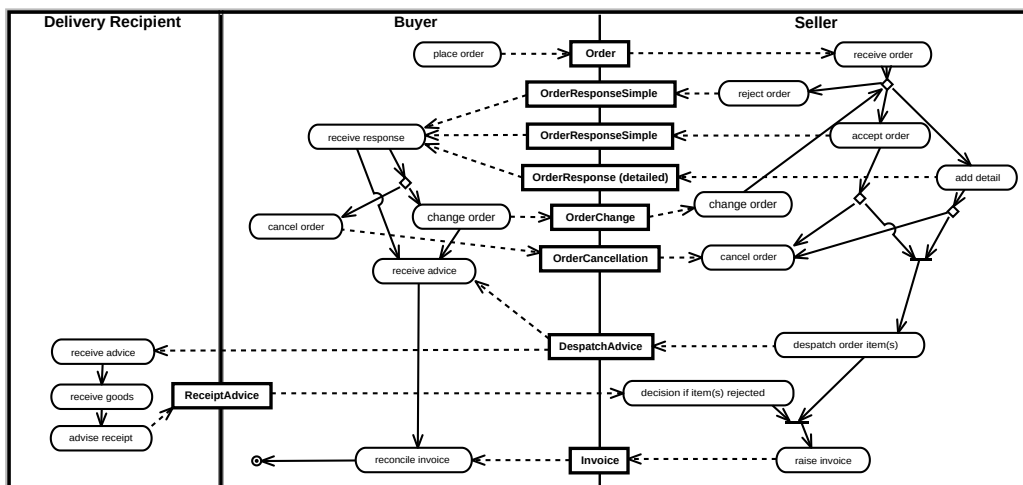
UBL-2.2-Tender-GuaranteeDeposit: 6 nodes, 5 flows. Labels print at about 9.7 pt.



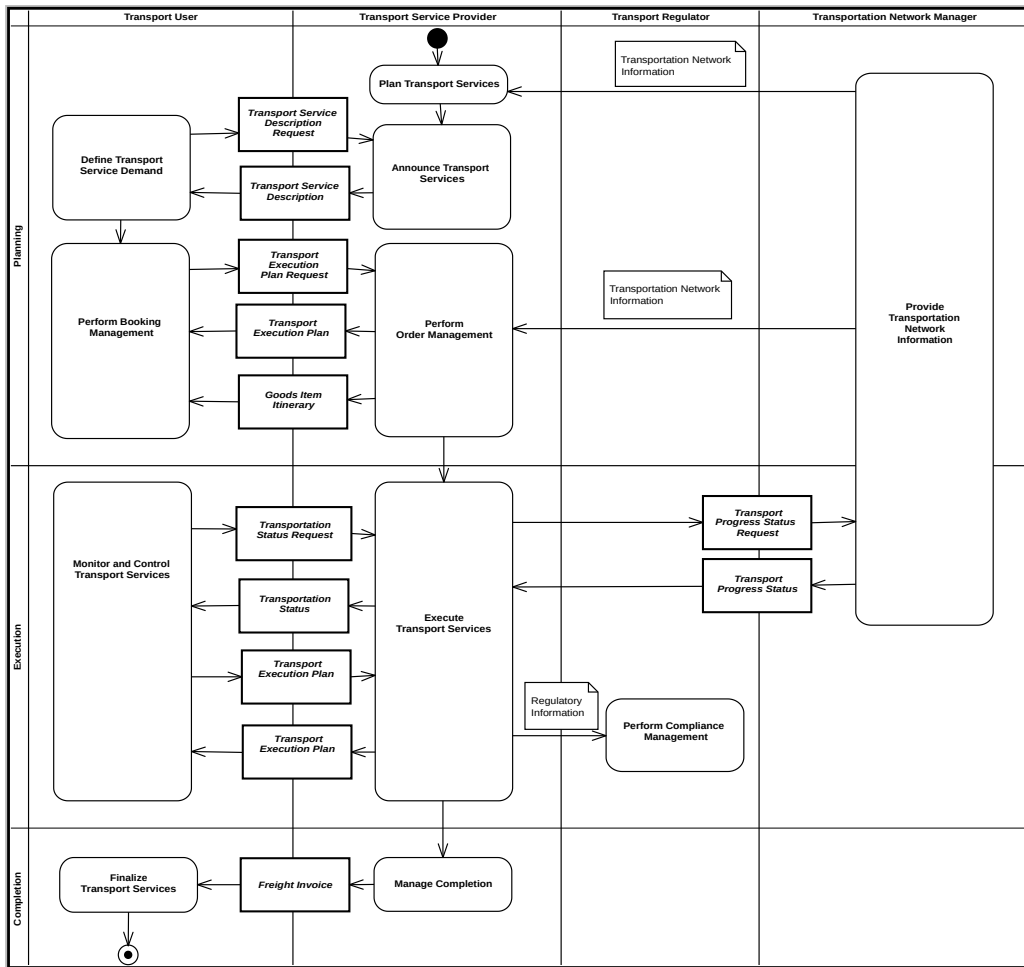
UBL-2.5-BillingwithDebitNoteProcess: 17 nodes, 20 flows. Labels print at about 4.5 pt.



UBL-2.2-CPFR-CreateOrderForecast: 18 nodes, 20 flows. Labels print at about 5.4 pt.

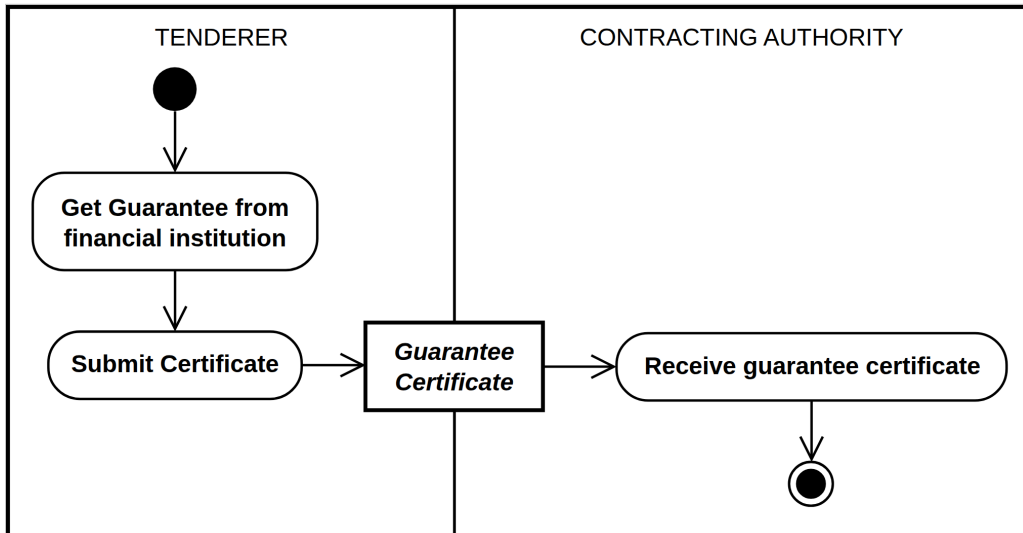


UBL-1.0-ProcurementProcess: 34 nodes, 43 flows. Labels print at about 3.9 pt.

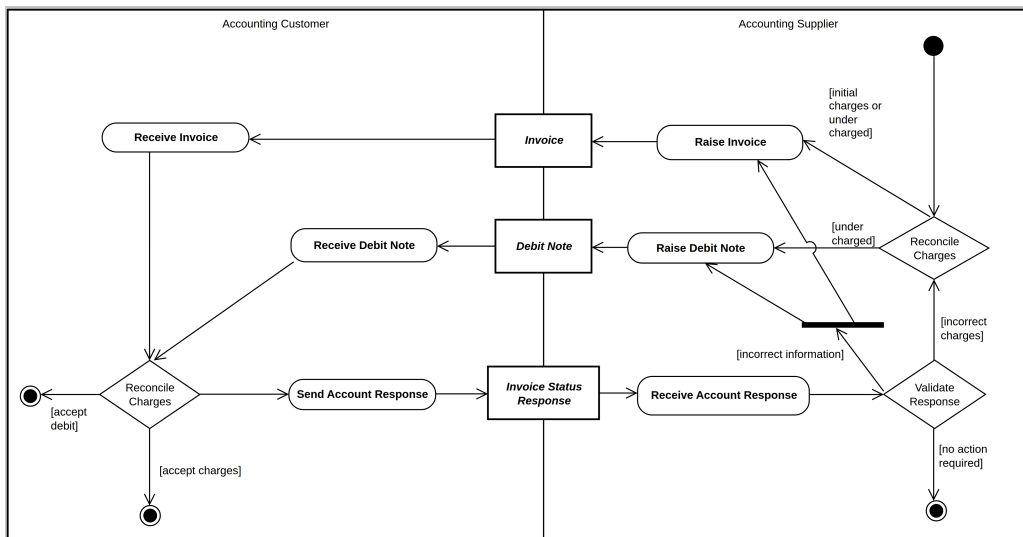


UBL-2.2-IMFM-IntermodalFreightManagementProcess: 28 nodes, 33 flows. Labels print at about 3.7 pt.

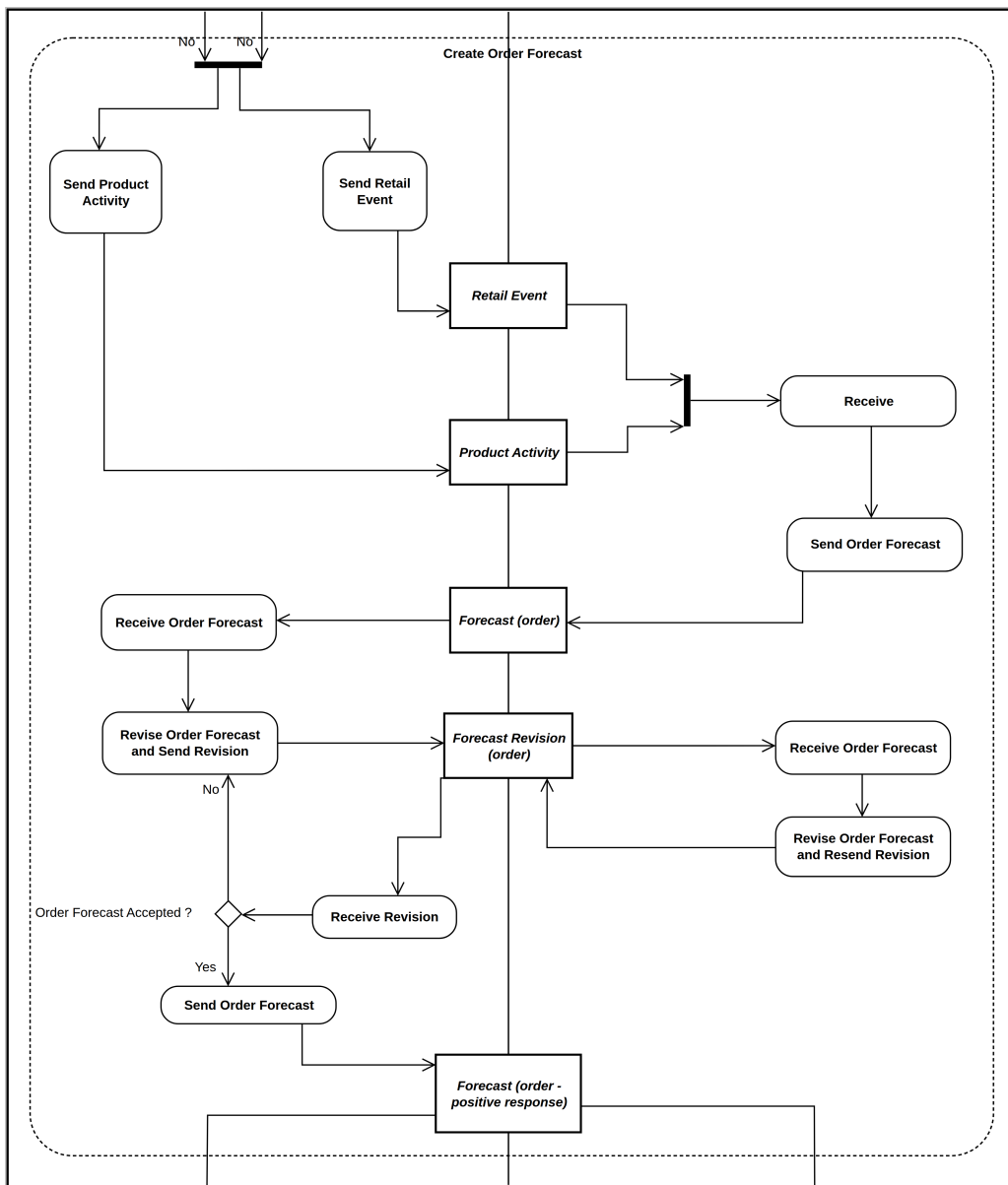
2. House style, each fitted to the column's width



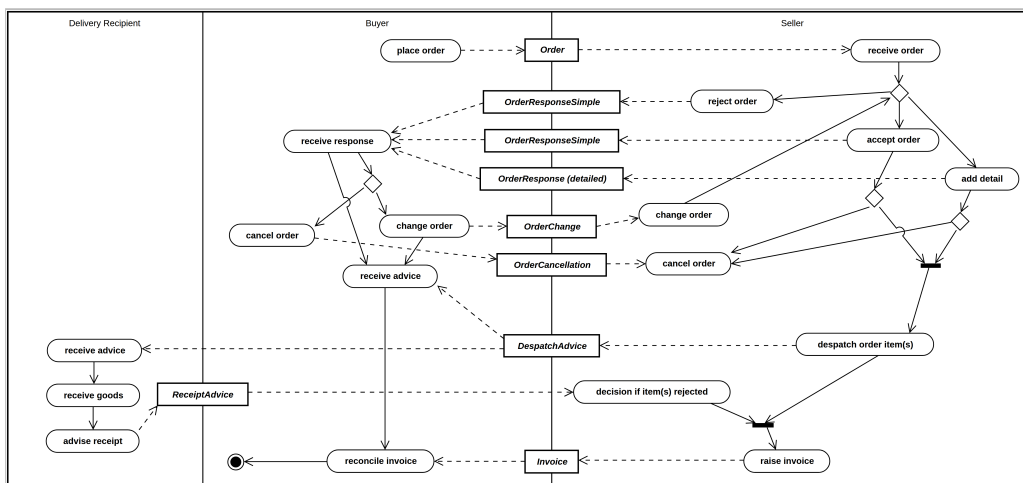
UBL-2.2-Tender-GuaranteeDeposit: labels print at 9.7 pt.



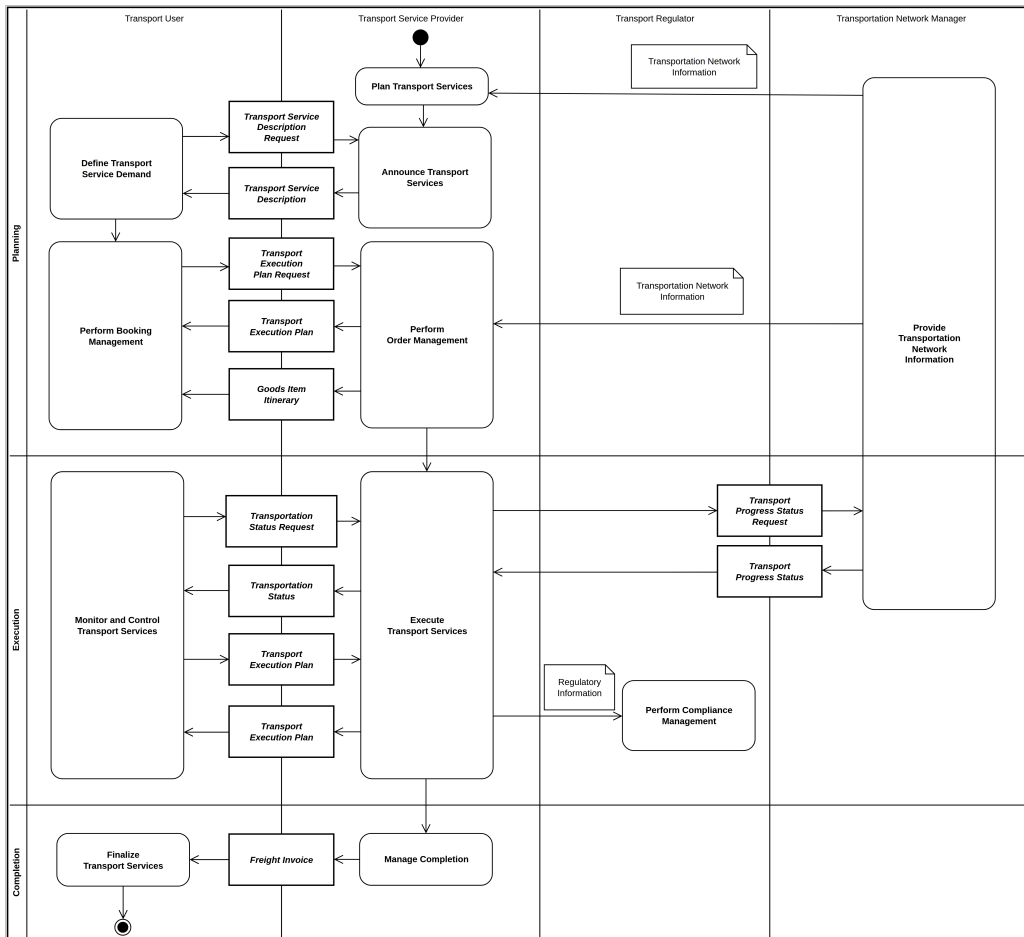
UBL-2.5-BillingwithDebitNoteProcess: labels print at 4.5 pt.



UBL-2.2-CPFR-CreateOrderForecast: labels print at 5.3 pt.

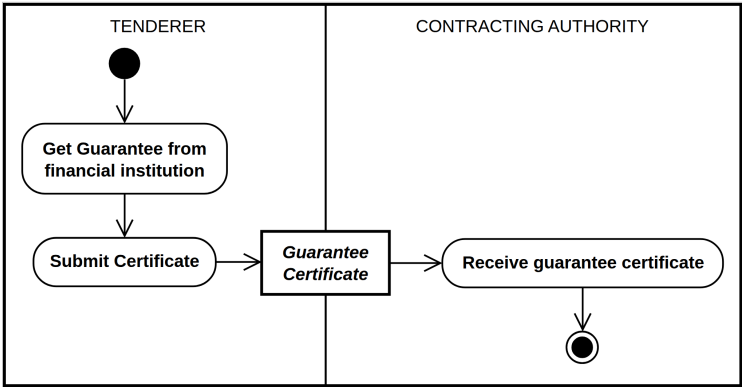


UBL-1.0-ProcurementProcess: labels print at 3.5 pt.



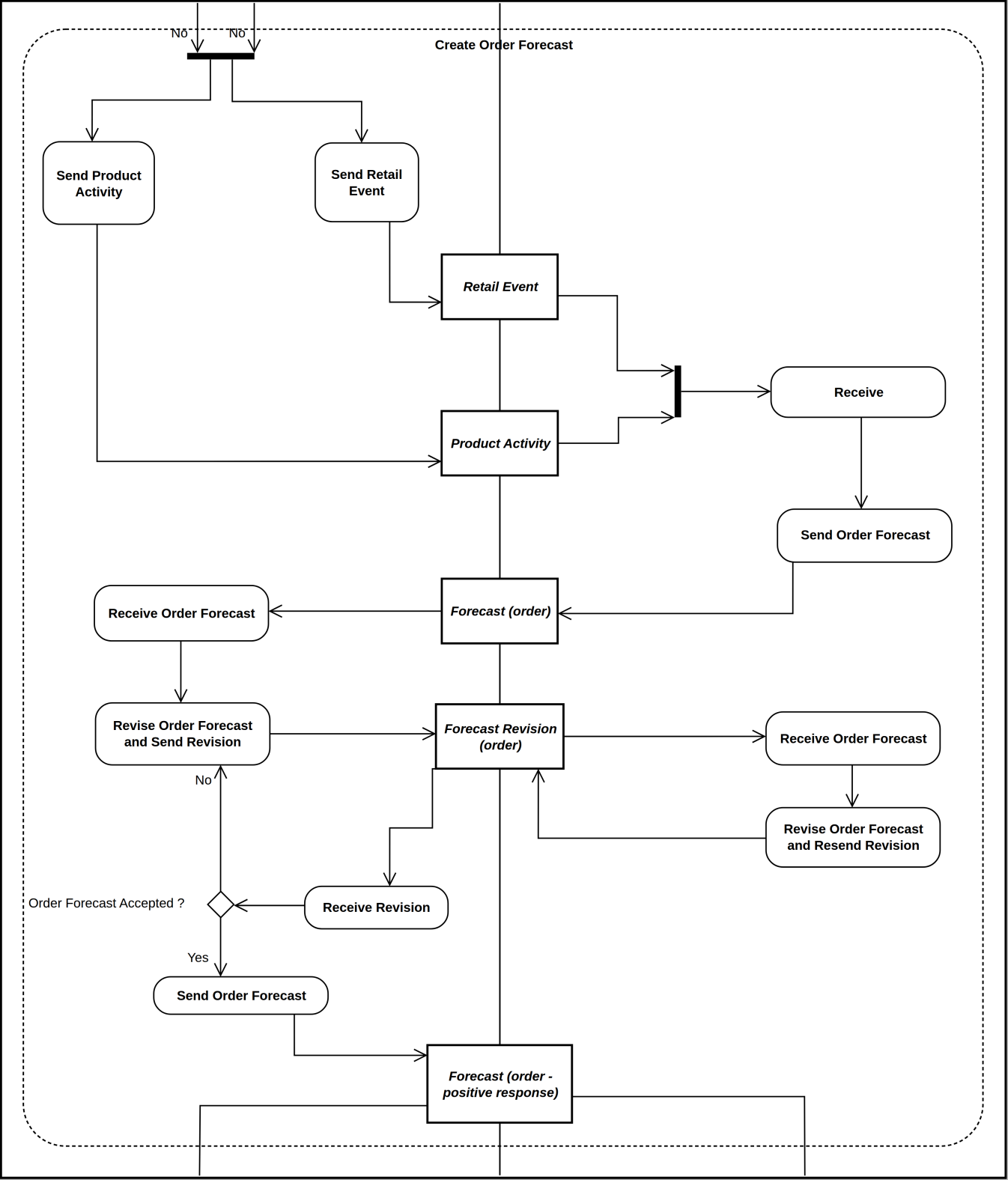
UBL-2.2-IMFM-IntermodalFreightManagementProcess: labels print at 3.6 pt.

3. House style at one scale: labels 7 pt on paper in every figure

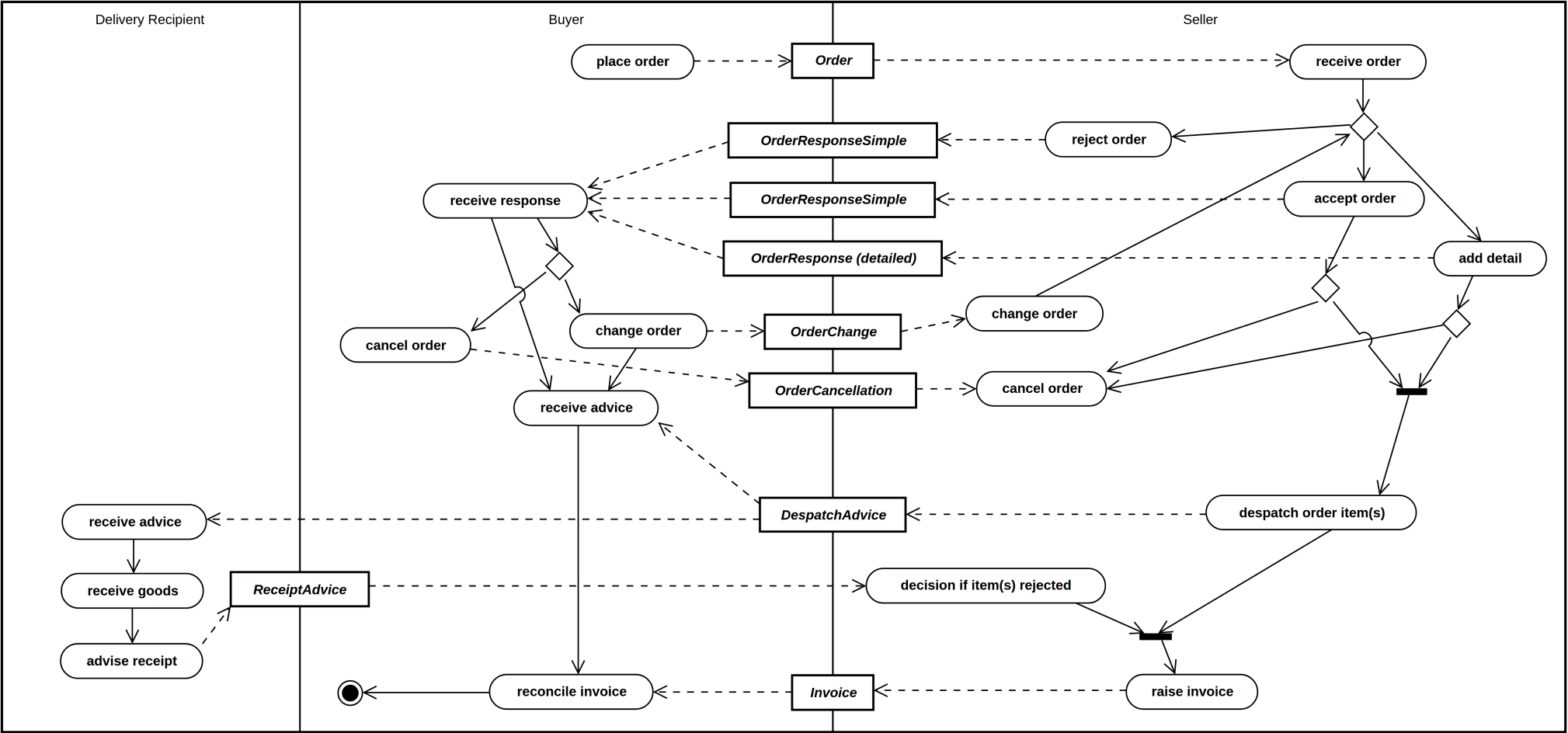


UBL-2.2-Tender-GuaranteeDeposit: 104 × 54 mm; needs A4, within the column.

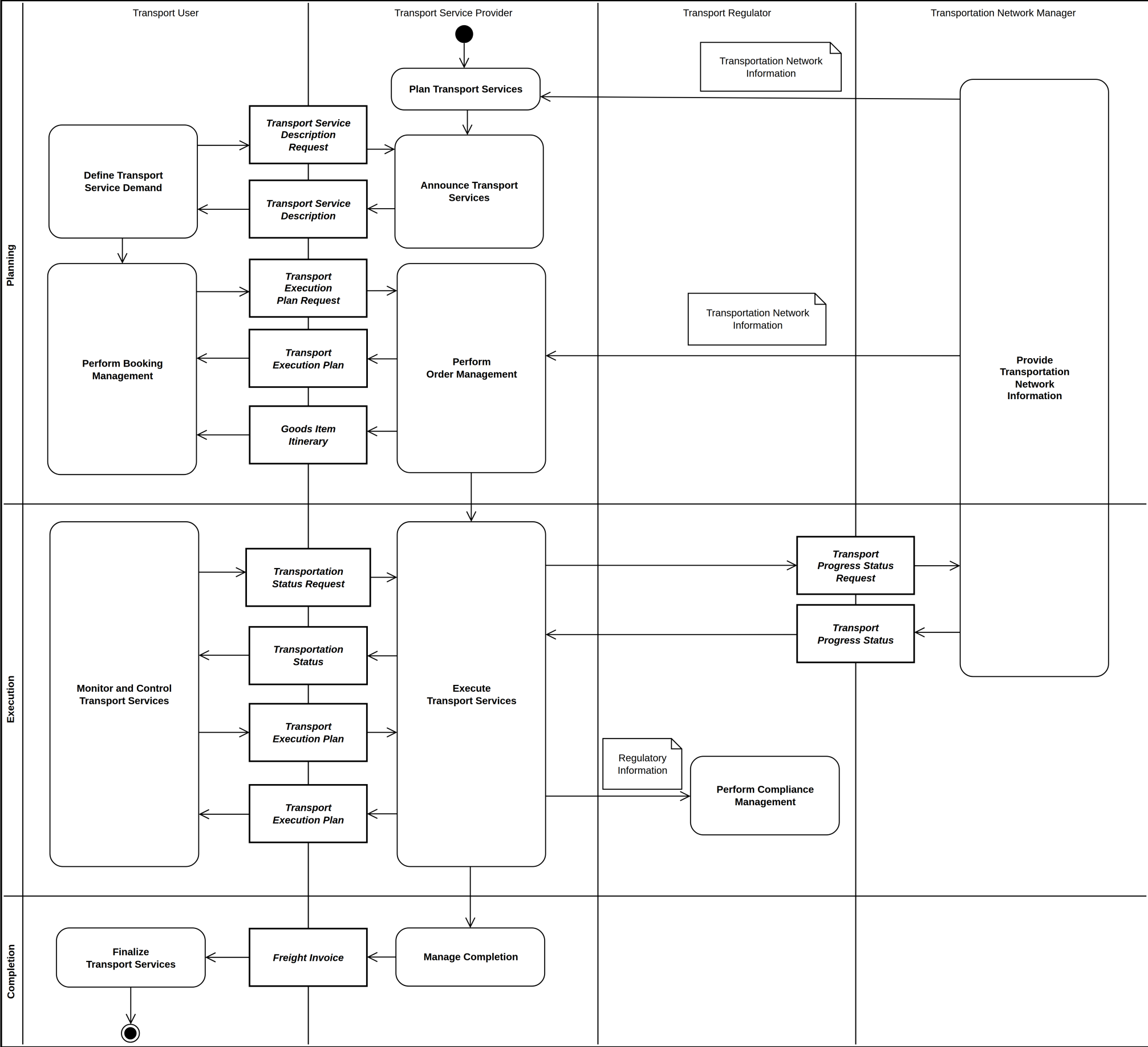




UBL-2.2-CPFR-CreateOrderForecast: 190 × 223 mm; needs an A3 landscape page.



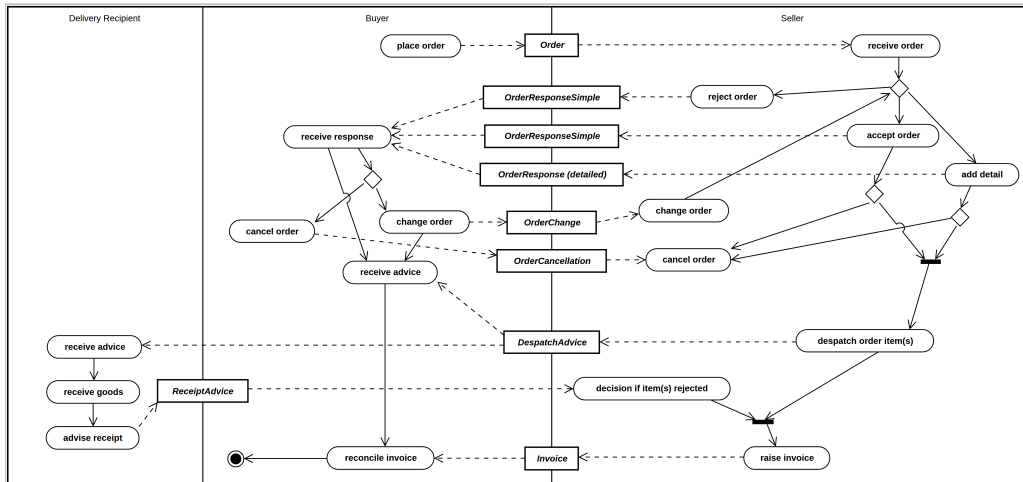
UBL-1.0-ProcurementProcess: 286 × 134 mm; needs an A3 landscape page.



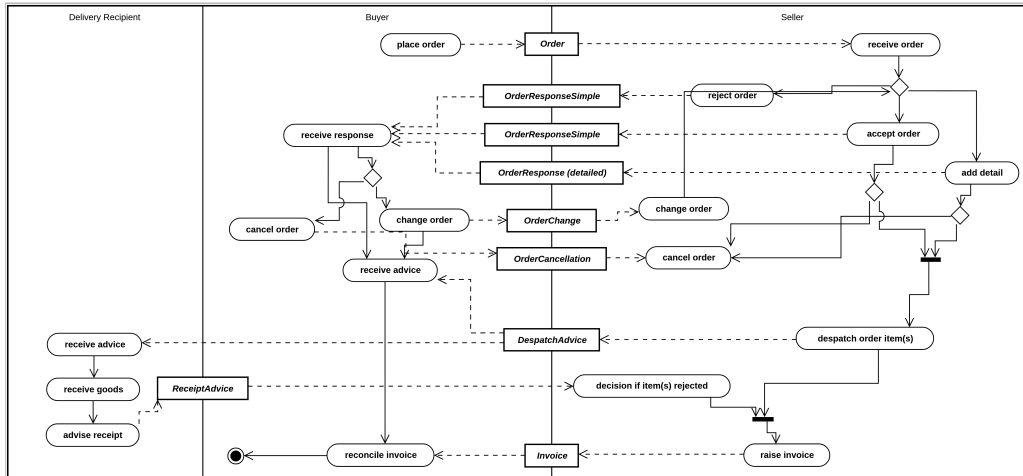
UBL-2.2-IMFM-IntermodalFreightManagementProcess: 286 × 261 mm; needs an A3 page.

4. The routing of flows: UBL-1.0-ProcurementProcess three ways

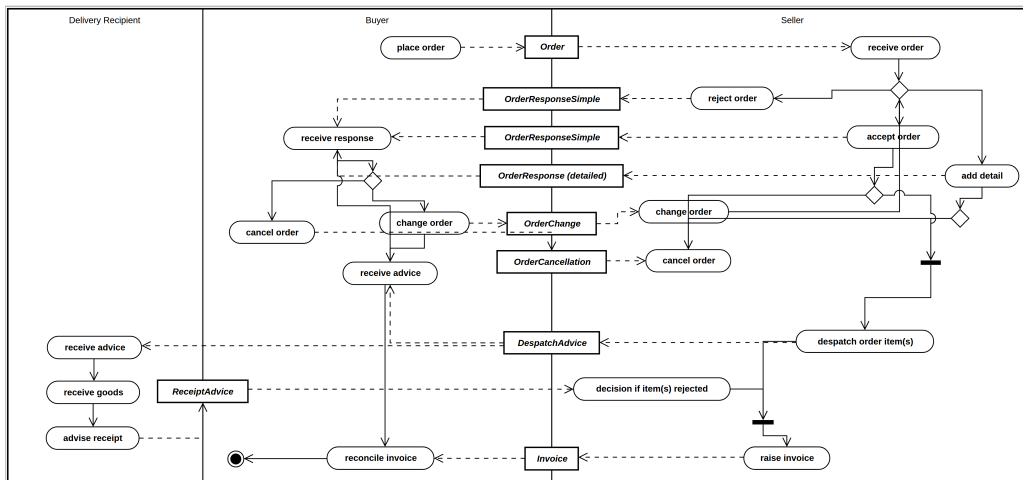
Half the figures mix flows at an angle with flows across and down. Could the house style make them all run at right angles? Here the angled flows of the densest figure, first as drawn, then re-routed.



As drawn (house style): flows at an angle kept, the ones almost level or upright made exactly so.



Right angles, simple: each angled flow as an L or a Z between the same points on the same sides.

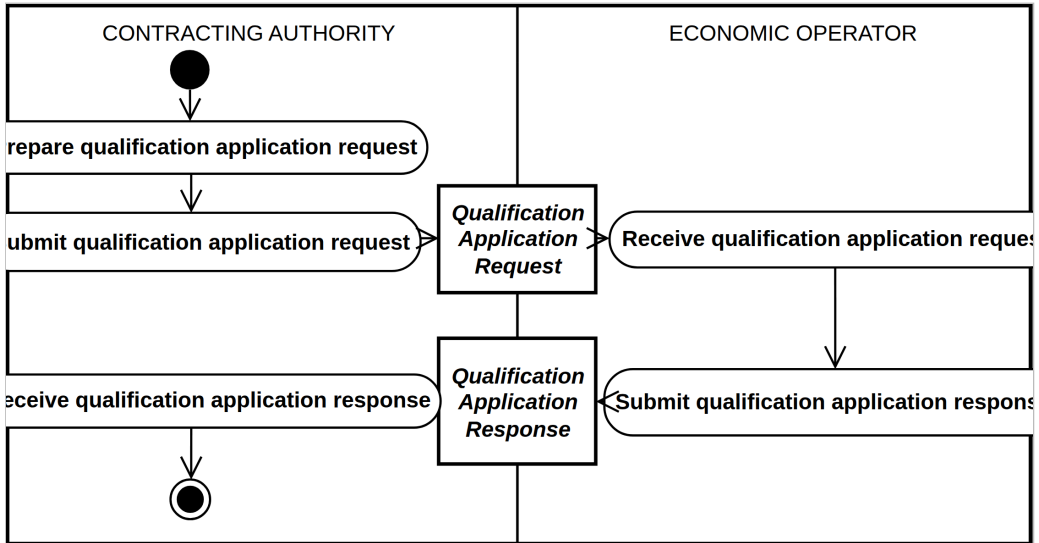


Right angles, draw.io's own router: draw.io chooses the sides and the route.

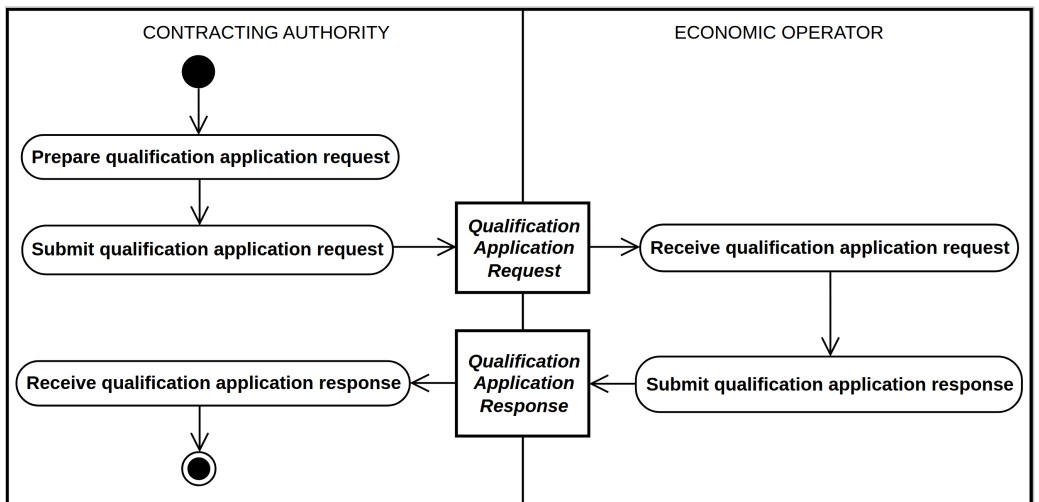
Neither re-routing reads better than the flows as drawn: routes run through boxes and over each other. These figures were laid out for angled flows; right angles would need each of them laid out anew.

5. Making room

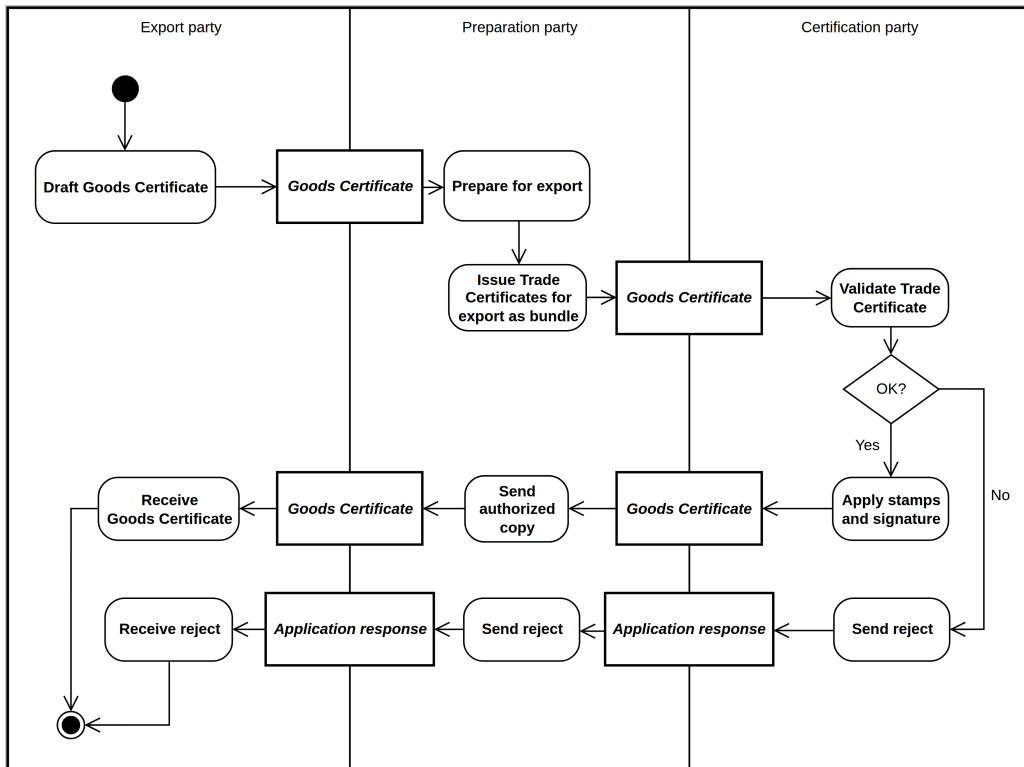
In the house style some boxes grow to fit their words, and in tightly drawn figures a flow is then too short for its arrowhead, or a box runs out of its lane. The figure is opened up where that happens: widened or lengthened at a line between the two, everything beyond moving along, so rows, columns and lanes stay aligned. Before, and after.



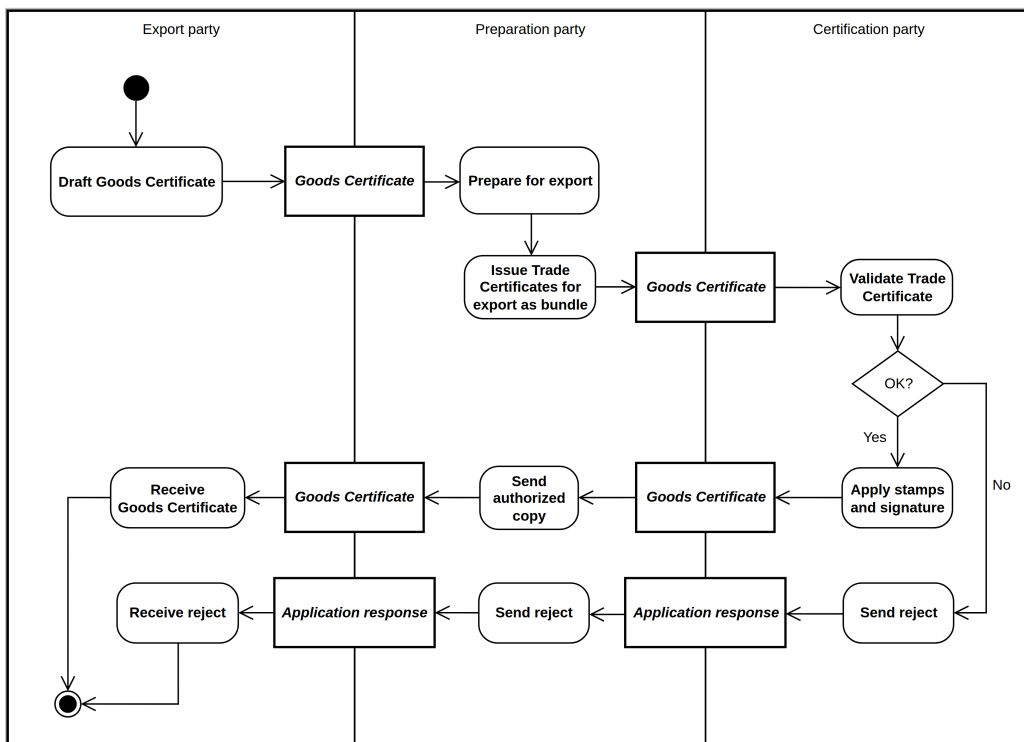
UBL-2.2-Tender-QualificationApplication: before.



UBL-2.2-Tender-QualificationApplication: after.



UBL-2.3-GoodsCertificateExportProcess: before.



UBL-2.3-GoodsCertificateExportProcess: after.